

ds30 Loader
Native console manual



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Introduction

ds30 Loader

ds30 Loader is a boot loader supporting PIC12, PIC16, PIC18, PIC24, dsPIC and PIC32 families of MCUs from Microchip. It supports all devices in each family out of the box (those in production). The firmware is written in assembler. The PC clients run on Windows, Linux, and macOS.

Prerequisites and Requirements

None.

Trademarks

All rights to copyrights, registered trademarks, and trademarks reside with their respective owners.

Usage

Installation

The ds30 Loader console does not require to be installed; it can be run directly from the bin director.

Usage syntax

The syntax is:

```
ds30LoaderNativeConsoleXXX options
```

Option values containing spaces need to be enclosed with quotation marks, the quotation marks should surround the entire argument like this "option=value".

Options summary

When unknown option or illegal option value is detected, the download is aborted. For more information about each command refer to the ds30 Loader main manual.

Short option name	Long option name	Description
Required		
-f ...	--file ...	Hex file
-k ...	--port ...	Port name
-r ...	--baudrate ...	Baud rate
-d ...	--device ...	Device name
Misc		
-h	--help	Display help
-l	--debugmode	Enable debug mode
n/a	--comm	[uart/socketcan]
Basic		
-p	--writef	Write flash
n/a	--writeaf	Write aux/boot flash
-e	--writee	Write EEPROM
n/a	--readf	Read main flash
n/a	--readaf	Read aux/boot flash
n/a	--reade	Read EEPROM
CAN		
n/a	--pic-id ...	CAN id of device
n/a	--ds30-id ...	CAN id of the ds30 Loader application
n/a	--ext	CAN extended frames
n/a	--dlc ...	CAN transmit data length code
n/a	--canfd	CAN FD
Advanced		
n/a	--disable-empty-pages	Empty pages will no be written
n/a	--blplp ...	Boot loader placement, pages from the end
n/a	--blsizep ...	Boot loader size in pages
n/a	--no-app-vector	Do not set the application vector
n/a	--no-bl-veector	Do not modify the reset vector

n/a	--echo-ver	Firmware echoes all received bytes
-s	--allow-bl-overwrite	Continue operation even if hex file contains data that will overwrite the boot loader
-x	--auto-baud	Send auto baud rate synchronization character
Timing		
n/a	--ht ...	Time to keep sendin hello before aborting [ms]
-a ...	--polltime ...	Polltime [ms]
-t ...	--timeout ...	Time before timeout [ms]
n/a	--port-open-delay ...	Delay after port is opened [ms]
Device reset		
-q ...	--reset-command ...	Reset device by specified command, this option may need to be enclosed in " under Linux.
-u ...	--reset-baud-rate ...	Baudrate for reset command
-b ...	--reset-time ...	Resettime [ms]
n/a	--reset-dlc xxx	CAN DLC for reset frames
n/a	--reset-ext	Send CAN extended frames
n/a	--reset-pic-id	CAN id of device to reset
Security		
n/a	--pw ...	Password to send to ds30 Secure Loader.

Return value

0 is returned after a successful operation, else -1.

Example

```
ds30LoaderNativeConsoleWinx86.exe --port COM12 --baudrate 115200 --  
debugmode --writef --writee --file "hex_file.hex"
```

Appendix A – Links

ds30 Loader website

<https://ds30loader.com>

ds30 Loader free edition website

<https://picbootloader.com>